Dataset Facts

Dataset Open Air Chicago Day Aggregations	
Original dataset	121,606 rows; 46 columns
Analysis sample	119,859 rows; 15 columns
Motivation	
Project leads	CDPH and UIC School of Public Health
Purpose	Community air-quality monitoring
Grant 1	NIST 60NANB23D206 (2023–2026)
Grant 2	ComEd / Exelon / RHP (2025–2026)
Composition	
Observation unit	One sensor per UTC day
Period	Jul 26, 2025 – Oct 26, 2026
Coverage	All 77 Chicago community areas
Main pollutants	PM2.5 and NO2
Rows excluded	1,747 with missing selected values
Collection	
Network	277 deployed sensors
Additional sites	Federal monitoring-site sensors
Measurement method	Clarity air-quality sensors
Cleaning and Labeling	
Quality control	Automated sensor data checks
Calibration	Raw and cloud-calibrated values
Analysis choice	Calibrated values; PM2.5 mass
Aggregation	Complete UTC-day averages
Uses and Distribution	
Uses	Neighborhood comparisons and trends
Access	Public CSV and API
Reuse	Processing and redistribution permitted
License	No specific license identified
Maintenance and Evolution	
Maintainers	CDPH and UIC School of Public Health
Update schedule	Daily; previous-day statistics
Snapshot retrieved	Oct 27, 2026
Limitations Non-regulatory sensors and daily averages cannot establish an individual's exact exposure; locations are sampled.	
Source: Chicago Data Portal (rtmx-vkjr); counts from student API analysis.	

Open Air Chicago Day Aggregations nutrition label. API snapshot retrieved October 27, 2026.

Original dataset: Chicago Data Portal, Open Air Chicago Day Aggregations.

Related metadata: Data.gov record.

Snapshot: Retrieved October 27, 2026. The original API extract contains 121,606 rows and 46 columns; the analysis sample retains 119,859 rows and 15 columns after removing missing values in selected fields.

Reuse: The student verified download, processing, and redistribution permission; no specific dataset license was identified.

Funding: Dr. Serap Erdal, Community-Driven Air Quality and Environmental Justice Assessment,

NIST grant 60NANB23D206 (Oct 2023–Sep 2026); ComEd Hyperlocal Air Quality Assessment, RHP Risk Management / ComEd / Exelon LLC (Jan 2025–Jan 2026).